

Assignment: Reproducibility and Extension Challenge

Objective

Reproduce key experiments from one machine learning research paper from the list below and/or extend the results in a novel way. Present your findings in a PowerPoint presentation.

Deliverables

- A PowerPoint presentation demonstrating your understanding, reproduction, and extension of the chosen paper
- A GitHub repository containing the implementation of the paper and any extensions
- Attendance at an individual group session after the group presentation to discuss your personal contributions and answer questions from the course instructor

Group Work

Students will form groups of three to four members. Each group will select one paper from the provided list. In general, each group should choose a single paper, although exceptions may be made. Students may also propose a paper, subject to instructor approval.

Presentation Guidelines

Duration: 20 minutes

Structure

Introduction

Briefly introduce the paper and its key contributions.

Understanding

Demonstrate a clear understanding of the paper. Explain the main concepts, methods, and results.

Reproduction

Present the reproduction of the paper's key experiments. Include the code used, datasets, and any challenges encountered.

Extension

Present any novel extensions or additional experiments you conducted. Explain the motivation for these extensions and discuss the outcomes.

Conclusion

Summarize your findings and comment on the reproducibility of the original results. Reflect on the value of your extensions and suggest possible future work.

Individual Group Sessions

Duration: 15 minutes per group

Purpose

During the individual session, each student should be able to:

- explain their personal contribution to the project,
- demonstrate understanding of the overall group work,
- answer questions related to the reproduction and extension tasks.

Advice on Experiment Reproduction

If reproducing the full experiments takes too long, you may use a representative subset of the original dataset to demonstrate the main principles of the paper. This can save time while still illustrating the key concepts and methods effectively. Make sure the chosen subset is representative enough to provide meaningful insights that are reasonably comparable to results on the full dataset.

Important Notes

Code Submission

All code must be made available in a public GitHub repository. Include a link to the repository in the presentation.

Use of External Code

If you use publicly available code, clearly acknowledge it and document any modifications you

make. More substantial extensions are expected if significant portions of the implementation are reused.

Timeline

Presentations will take place on **18 and 19 June**.

The written exam will take place, during the last week of June.

List of Papers

Neural Machine Translation (NMT)

- **Neural Machine Translation by Jointly Learning to Align and Translate**
Paper: arXiv:1409.0473
- **Neural Machine Translation in Linear Time**
Paper: arXiv:1610.10099

Question Answering (QA)

- **Dynamic Coattention Networks for Question Answering**
Paper: arXiv:1611.01604
- **Improving Coreference Resolution by Learning Entity-Level Distributed Representations**
Paper: arXiv:1606.01323

Word Embeddings

- **Learned in Translation: Contextualized Word Vectors**
Paper: arXiv:1708.00107

Recent Papers (Transformers, RAG, LoRA, etc.)

- **DistilBERT: a distilled version of BERT: smaller, faster, cheaper and lighter**
Paper: arXiv:1910.01108
- **BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding**
Paper: arXiv:1810.04805
- **LoRA: Low-Rank Adaptation of Large Language Models**
Paper: arXiv:2106.09685
- **Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks**
Paper: arXiv:2005.11401

- **TinyStories: How Small Can Language Models Be and Still Speak Coherent English?**
Paper: arXiv
- **Chain-of-Thought Prompting Elicits Reasoning in Large Language Models**
Paper: arXiv
- **Q8BERT: Quantized 8-Bit BERT**
Paper: arXiv
- **Low-Rank Few-Shot Adaptation of Vision-Language Models**
Paper: arXiv:2405.18541
- **TinyBERT**
Paper: arXiv:1909.10351
- **ALBERT**
Paper: arXiv:1909.11942